



ENGINEERING MATHEMATICS - II

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UNIT 5 : Fourier series

Session : 3

Sub Topic : Problems on Fourier Series

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Recall!!

We know that, for a periodic function $f(x)$ of period 2π , defined over $(-\pi, \pi)$, the Fourier series expansion is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \dots \dots \dots (1)$$

where $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx.$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx, \quad n = 1, 2, 3 \dots$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx, \quad n = 1, 2, 3 \dots$$

Problems on Fourier series

1. Find the Fourier series of $f(x) = x + x^2$ in $(-\pi, \pi)$. Hence deduce that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

Solution:

Given $f(x) = x + x^2$

We have $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \dots \dots (1)$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$
$$= \frac{1}{\pi} \int_{-\pi}^{\pi} x + x^2 dx$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x^2 dx \quad (\text{since } \int_{-\pi}^{\pi} x dx = 0 \text{ is an odd function in the interval } (-\pi, \pi))$$

$$a_0 = \frac{2}{\pi} \left(\frac{x^3}{3} \right)_0^{\pi} = \frac{2\pi^2}{3}$$

Solution contd...

$$\begin{aligned}a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \\&= \frac{1}{\pi} \int_{-\pi}^{\pi} (x + x^2) \cos nx dx \\a_n &= \frac{1}{\pi} \left[(x + x^2) \left(\frac{\sin nx}{n} \right) - (1 + 2x) \left(\frac{-\cos nx}{n^2} \right) + (2) \left(\frac{-\sin nx}{n^3} \right) \right]_{-\pi}^{\pi} \\a_n &= \frac{1}{\pi} \left[(1 + 2\pi) \left(\frac{\cos n\pi}{n^2} \right) - (1 - 2\pi) \left(\frac{\cos n\pi}{n^2} \right) \right] \\&= \frac{4}{n^2} \cos n\pi \\a_n &= \frac{4}{n^2} (-1)^n\end{aligned}$$

Solution contd...

$$\begin{aligned}b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx \\&= \frac{1}{\pi} \int_{-\pi}^{\pi} (x + x^2) \sin nx dx \\&= \left[(x + x^2) \left(\frac{-\cos nx}{n} \right) - (1 + 2x) \left(\frac{-\sin nx}{n^2} \right) + 2 \frac{\cos nx}{n^3} \right]_{-\pi}^{\pi} \\b_n &= \frac{1}{\pi} \left[(\pi + \pi^2) \left(\frac{-\cos n\pi}{n} \right) - (-\pi + \pi^2) \left(\frac{-\cos n\pi}{n} \right) + 2 \frac{\cos n\pi}{n^3} - 2 \frac{\cos n\pi}{n^3} \right] \\&= \frac{-2}{n} \cos n\pi \\b_n &= \frac{-2}{n} (-1)^n\end{aligned}$$

Solution contd...

∴ (1) becomes,

$$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} (-1)^n \cos nx + \frac{-2}{n} (-1)^n \sin nx \right) \dots\dots\dots (2)$$

This is the required Fourier series

Deduction:

Put $x = \pi$.

Since the function is not defined at π , $f(x)$ at $x = \pi$ is given by

$$\begin{aligned} f(x) &= \frac{(f(\pi+) + f(\pi-))}{2} \\ &= \frac{1}{2} [\pi + \pi^2 - \pi + \pi^2] \end{aligned}$$

Solution contd...

$$f(x) = \pi^2$$

$\therefore$ (2) becomes

$$\pi^2 = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} \right)$$

$$\frac{\pi^2}{6} = \sum_{n=1}^{\infty} \left(\frac{1}{n^2} \right)$$



THANK YOU

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